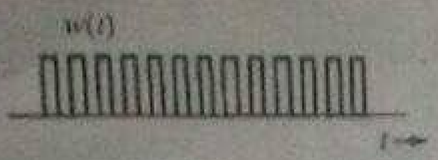


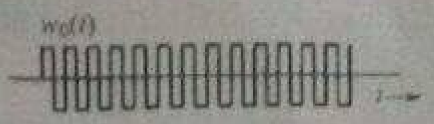


Figure 2 (a) Figure 2(b)

onsider a VSB amplitude modulation system. The baseband signal is an audio signal of bandwidth 4kHz. The arrier frequency $f_c = 1500kHz$. Suppose the transmission vestigial filter $H_i(\omega)$ has a frequency response as hown in Figure 3 below

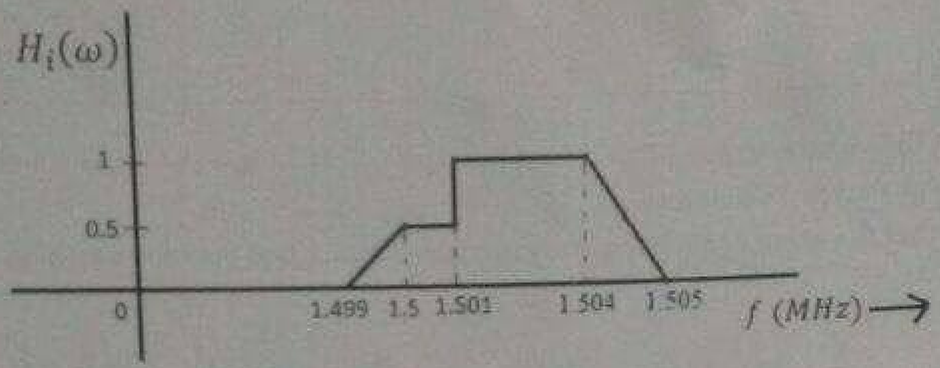


Figure 3

1) Design and illustrate a receiver system block diagram (synchronous detector).
2) Find the bandwidth of this transmission.
3) Describe and sketch the necessary equalizer filter response $H_o(\omega)$ for distortion-less reception.

uestion No. 4 Marks [5+15]

) Draw and explain the composite baseband spectrum for stereophonic FM broadcasting.

) An angle-modulated signal with carrier frequency $\omega_c = 2\pi \times 10^6$ is described by the equation

$$\varphi_{EM}(t) = 5\cos(\omega_c t + 20\sin 1000\pi t + 10\sin 2000\pi t)$$

1) Find the power of the modulated signal.
2) Find the frequency deviation Δf and the phase deviation $\Delta\varphi$.
3) Estimate the bandwidth of $\varphi_{EM}(t)$.

Question No. 5 Marks [5+15]

a) Show that a Delta demodulator is just an accumulator, assuming zero initial condition $m_q[0]=0$.

b) Five sensor output signals, each of bandwidth 240 Hz, are to be transmitted simultaneously by binary PCM The signals must be sampled at least 30% above the Nyquist rate. Framing and synchronization requires a additional 1.5% extra bits. The PCM encoder uses a non-uniform quantizer with $\mu = 100$ to convert thes signals before they are time-multiplexed into a single data stream. The required SNR is to be at least 43dB i this case.

1) Find the minimum number of bits required to code the non-uniform quantizer.
2) Determine the minimum possible data rate (bits per second) that must be transmitted
3) Determine the minimum bandwidth required to transmit the multiplexed signal.

$$\frac{3L^2}{[\ln(1+4)]^2}$$

National University of Computer and Emerging Sciences, Lahore Campus

	Course:	Analog and Digital Communications	Course Code:	EE323
	Program:	BS(Electrical Engineering)	Semester:	Fall 2016
	Duration:	3 Hours	Total Marks:	100
	Paper Date:	16-Dec-16	Weight	50%
	Section:	ALL	Page(s):	2
	Exam:	Final	Roll No:	

Instruction/Notes: Attempt all parts of a question together.

Question No. 1 Marks [

a) Derive that the signal energy E_g found from the signal $g(t)$ can also be determined from its Fourier tran $G(\omega)$ through Parseval's theorem.

b) Check whether the given signal $x(t)$ is energy or power signal, find the corresponding energy/power.

$$x(t) = (e^{-3t} + 1)u(t)$$

Question No. 2 Marks

a) If the Fourier Transform of a signal is enough to determine its Spectral Density, why do we complic lives by defining the autocorrelation functions?

b) The random binary signal $x(t)$ shown in Figure 1 transmits one digit every T_b seconds. A binary **1** is tran by a pulse $p(t)$ of width $T_b/2$ and amplitude B; a binary **0** is transmitted by no pulse. The digits **1** ar equally likely and occur randomly. Determine the autocorrelation function $\mathcal{R}_x(\tau)$ and the PSD $S_x(\omega)$

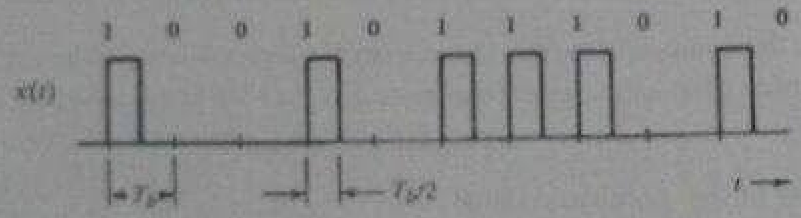


Figure 1

Question No. 3 Mar

a) The Fourier series for the square pulses $w(t)$ in Figure 2 (a) and $w_0(t)$ in Figure 2 (b) are as follows:

$$w(t) = \frac{1}{2} + \frac{2}{\pi}\left(\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \frac{1}{5}\cos 5\omega_c t - \cdots\right)$$

$$w_0(t) = \frac{4}{\pi}\left(\cos\omega_c t - \frac{1}{3}\cos 3\omega_c t + \frac{1}{5}\cos 5\omega_c t - \cdots\right)$$

Show working of a series-bridge diode modulator to accomplish DSB-SC with the help of its block diagra

Question # 3 [4+4+3+4] CLO3

For each of the following baseband signals:

(i) $m(t) = 2\text{Cos}(1000t)$(ii) $m(t) = 4\text{Cos}1000t + 2\text{Cos}2000t$

(iii) $m(t) = 2\text{Cos}(1000t)\text{Cos}(3000t)$

a) Sketch the spectrum of m(t)
b) Sketch the spectrum of DSB-SC signal m(t)*Cos(10,000t)
c) Identify USB and LSB spectra
d) Identify the frequencies in the baseband and the corresponding frequencies in the DSB-SC, USB and LSB spectra. Explain the nature of frequency shifting in each case.

Question # ~~3~~ 4 [5] CLO4

Determine the power efficiency (□) and the percentage of the total power carried by the sidebands of the AM wave for tone modulation when modulation index is 0.5 and 0.3.

Question # 5 [10+10] CLO5

a) An angle-modulated signal with carrier frequency $\omega c = 2\pi\times1000000$ is described by:

$$\varphi EM(t) = 10\cos(\omega ct + 0.1\sin(1000)\pi t)$$

i) Find the power of the modulated signal.
ii) Find the frequency deviation Δf.
iii) Find the phase deviation $\Delta\phi$.
iv) Estimate the bandwidth of $\varphi EM(t)$.
v) Considering $\varphi EM(t)$ as a PM signal with $kp=10$, find $m(t)$.
vi) Considering $\varphi EM(t)$ as an FM signal with $kf=10\pi$, find $m(t)$.

b) Design an Armstrong indirect FM modulator to generate an FM carrier with a carrier frequency of 96 MHz and Δf=20 kHz. A narrow-band FM generator with fc=200 kHz and adjustable Δf in the range of 9 to 10 Hz is available. The stock room also has an oscillator with adjustable frequency in the range of 9 to 10 MHz. There is a bandpass filter with any required center frequency, and only frequency doublers are available.

School of Electrical Engineering

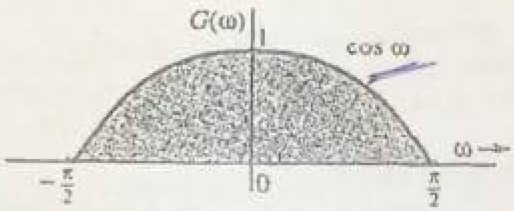
National University of Computer and Emerging Sciences, Lahore Campus

Course:	Analog and Digital Communication	Course Code:	EE323
Program:	BS(Electrical Engineering)	Semester:	Spring 2017
Duration:	60 Minutes	Total Marks:	30
Paper Date:	21st February, 2017	Weight	15%
Section:	EE	Page(s):	1
Exam:	Midterm-1	Roll No:	

Instruction/Notes:

Question # 1 [6+4 = 10 marks] CLO 1

(a) Find the inverse Fourier transform of the following waveform using definition.



(b) The spectrum $G(w)$ is shifted in frequency by $\pm w_o$, plot $G(w - w_o)$ and $G(w + w_o)$ and find the inverse Fourier transform of the new specturm by using some property.

Question # 2 [10 marks] CLO 2

For the following Gaussian pulse (widely used in communication modeling), verify Parseval's theorem by first finding the energy of the pulse by using $x(t)$.

$$x(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2\sigma^2}}$$

2σ√π

$$Hint: \text{Use the fact that } \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$\frac{\omega^2\sigma^2}{2}$

Question # 3 [6+4 = 10 marks] CLO 3

Consider the following sinusoidal message $m(t)$ and carrier $c(t)$signals,

$$m(t) = A_m \cos \omega_m t \text{ and } c(t) = A_c \cos \omega_c t, \qquad where\ \omega_c \gg \omega_m$$

(a) Write down the expression for DSB-SC modulated signal, $g_{DSB-SC}(t)$. Also plot the spectra of the modulated signal $g_{DSB-SC}(t)$. Identify USB and LSB of the spectrum, $G_{DSB-SC}(\omega)$. For the message signal in hand, name the type of modulation.

(b) $g_{DSB-SC}(t)$ is demodulated using a coherent detection. Draw a labeled block diagram of the coherent detection with proper assigning of voltages before and after each block.

6

6 [4+6] CLO6

e and prove the Sampling theorem.

A signal $g(t) = sinc^2(5\pi t)$ is sampled (using uniformly spaced impulses) at a rate of (i) 5 Hz (ii) 10 Hz (iii) 20 Hz . For each of the three case:

1) Sketch the sampled signal
2) Sketch the spectrum of the sampled signal
3) Explain whether you can recover the signal g(t) from the sampled signal.
4) If the sampled signal is passed through and ideal low-pass filter of bandwidth 5 Hz, sketch the spectrum of the output signal.

25

Question # 7 [3+4+4+4] CLO7 15

A compact disc (CD) records audio signals digitally by using PCM. Assume the audio signal bandwidth to be 15 kHz.

a) What is the Nyquist rate?
b) If the Nyquist samples are quantized into L=65,536 levels and then binary coded, determine the number of binary digits required to encode a sample.
c) Determine the number of binary digits per second (bits/s) required to encode the audio signal.
d) For practical reasons discussed in the text, signals are sampled at a rate well above the Nyquist rate. Practical CDs use 44,100 samples per second. If L=65,536, determine the number of bits per second required to encode the signal, and the minimum bandwidth required to transmit the encoded signal.

90m

128 x64 0.96M 98

125 200 100 95

Course:	Analog and Digital Communication	Course Code:	EL323
Program:	BS(Electrical Engineering)	Semester:	Spring 2017
Duration:	180 Minutes	Total Marks:	100
Paper Date:	16-May-17	Weight	50%
Section:	EE	Page(s):	4
Exam:	Final Exam	Roll No.	

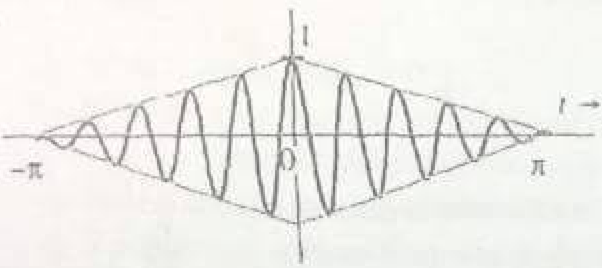
Instruction/Notes:

Question # 1 [9+6] CLO1

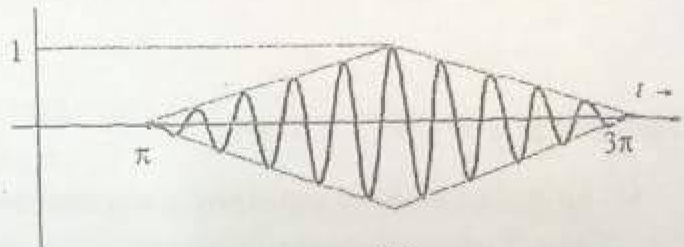
The signals in the following figure are modulated signals with carrier Cos(10t). $2\pi f = \omega$

a) Find the Fourier transform of these signals using the appropriate properties of the Fourier transform (use the table attached at the last page).

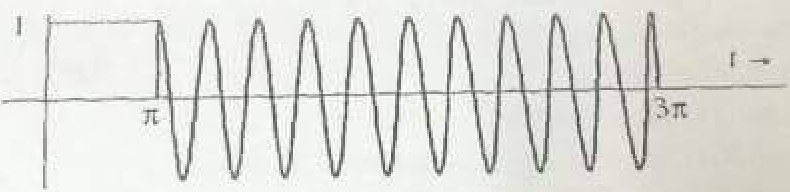
b) Sketch the amplitude and phase spectra for figure (a) and (b)



(a)



(b)



(c)

Question # 2 [10 + 10] CLO2

a) For the signal g(x) = Sinc(k.x) . find the energy of the signal using Parsevals's theorem.

b) Generalize the Parseval's theorem to show that for real, Fourier transformable signals $g_1(t)$ and $g_2(t)$:

$$\int_{-\infty}^{\infty} g_1(t)g_2(t)\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} G_1(-\omega)G_2(\omega)\,d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} G_1(\omega)G_2(-\omega)\,d\omega$$

Question # 2 [4+6+5 = 15 marks]

(a) Draw the block diagram structures for the Transmitter and Receiver for VSB transmission.

(b) The vestigial filter transfer function $H_i(\omega)$ of the transmitter module is shown below. The carrier frequency is $f_c = 10\ kHz$ and the baseband signal bandwidth is $4\ kHz$. Plot the corresponding transfer function of the equalizer filter $H_o(\omega)$ of the receiver module on the same scale as $H_i(\omega)$.

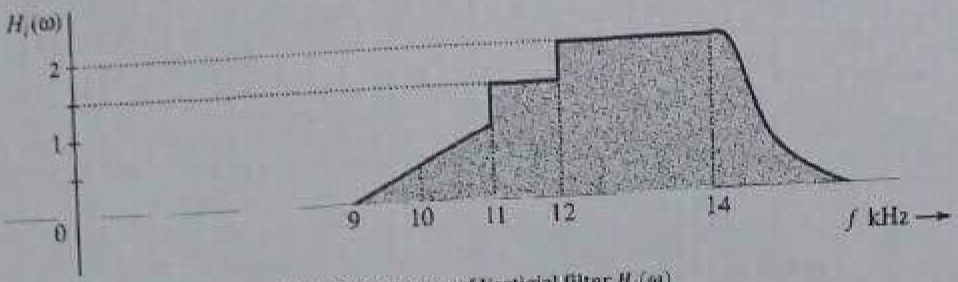


Figure 3 Spectrum of Vestigial filter $H_i(\omega)$

(c) Mathematically prove that the Hilbert Transform is equivalent to a 90 degrees phase shifter.

Question # 3 [8+7 = 15 marks]

(a) An angle-modulated signal with carrier frequency $\omega_c = 2\pi \times 10^6$ is described by

$$\varphi_{EM}(t) = 10\cos(\omega_c t + 0.1\sin(10^3)\pi t)$$

(i) Find the power of the modulated signal.

(ii) Find the frequency deviation Δf.

(iii) Find the phase deviation $\Delta\phi$.

(iv) Estimate the bandwidth of $\varphi_{EM}(t)$.

(v) Considering $\varphi_{EM}(t)$ as a PM signal with $k_p = 10$, find $m(t)$.

(vi) Considering $\varphi_{EM}(t)$ as an FM signal with $k_f = 10\pi$, find $m(t)$.

(b) Design an Armstrong indirect FM modulator to generate an FM carrier with a carrier frequency of 96 MHz and $\Delta f = 200\ kHz$. A narrow-band FM generator with $f_c = 200\ kHz$ and adjustable Δf in the range of 9 to 10 Hz is available. The stock room also has an oscillator with adjustable frequency in the range of 9 to 10 MHz. There is a bandpass filter with any required center frequency, and only frequency doublers are available.

Question # 4 [6+9 = 15 marks]

(a) State and prove the Sampling theorem.

(b) A compact disc (CD) recording system samples each of two stereo signals with a 16-bit analog-to-digital converter (ADC) at 44.1 kbps.

(i) Determine the output signal-to-quantizing-noise ratio for a full-scale sinusoid.

(ii) The bit stream of digitized data is augmented by the addition of error-correcting bits, clock extraction bits, and display and control bit fields. These additional bits represent 100 percent overhead. Determine the output bit rate of the CD recording system.

(iii) The CD can record an hour's worth of audio file. Determine the number of bits recorded on a CD.

Question # 5 [10+5 = 15 marks]

(a) A message signal $m(t)$ is transmitted by binary PCM without compression. If the SNR (signal-to-quantization-noise ratio) is required to be at least $\left(\frac{S_o}{N_o}\right)_{dB} = 50$ dB, determine the minimum value of L required for the following message signals. Also determine the SNR obtained with this minimum L for these message signals.

(i) $m(t)$ is sinusoidal with peak amplitude m_p.

(ii) $m(t)$ is triangular with peak amplitude m_p.

(b) For a binary PCM signal, determine L if the compression parameter $\mu = 100$ and the minimum SNR required is 50 dB. Determine the output SNR with this value of L.

Hint: $\frac{S_o}{N_o} = 3L^2\left(\frac{\overline{m^2(t)}}{m_p^2}\right)$ changes to $\frac{S_o}{N_o} = \frac{3L^2}{[\ln(1+\mu)]^2}$ for μ-law compandor.

Question # 6 [5+7+8 = 20 marks]

(a) Draw the **essential** block diagram of a typical digital communication system (DCS) starting from information source upto the sink with proper labeling.

(b) Explain the basis for deriving the likelihood ratio test

$$z(T) \gtrless_{H_2}^{H_1} \gamma \qquad \frac{q_1+q_2}{2}$$

and under what conditions the above test would be the optimum decision making criteria.

(c) A matched filter is a linear filter designed to provide the maximum SNR at its output for a given transmitted symbol waveform. SNR at sampling instant $t = T$, is

$$\left(\frac{S}{N}\right)_T = \frac{a_i^2}{\sigma_0^2} \quad \left\{\frac{2E}{N_0}\ Ed\right\}$$

Find the filter transfer function $H(f)$ and impulse response $h(t)$ of the filter that maximizes the above SNR equation. Also comment on the dependency of maximum output SNR.

PSD, not, energy

EE323 - Analog and Digital Communications
Midterm Exam-2 Fall 2015
National University of Computer and Emerging Sciences, Lahore
Course Code: EE323

EE323- Analog and Digital Communications

Midterm Exam-1 Fall 2015

National University of Computer and Emerging Sciences, Lahore

Course Code: EE323 — Semester: Fall 2015

September 2015 — Time: 60 minutes

Student Name Shehzina Ali — Roll number L 13-4432

Q.No.1

The time autocorrelation function of a random signal is given as:

$$R(\tau) = e^{-2\alpha|\tau|} \qquad -\infty < \tau < \infty$$

a) Find the power spectral density $S(\omega)$.

b) Compute the essential bandwidth of the signal that contains 90% of the total power.

You may use following formula where needed:

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$$

Q. No. 2

The relationship between channel capacity C (in bits per second, bps) and bandwidth B (in Hertz, Hz) in the presence of additive white Gaussian noise (AWGN) is highlighted by Shannon's equation:

$$C = B \log_2(1 + SNR) \quad bps$$

where SNR is signal to noise ratio, the ratio between the signal power and the noise power level. Suppose that the spectrum of the communication channel is between 10 MHz and 12 MHz, and the SNR requirement for the reliable communication in the presence of noise is 24 dB.

$$Hint: SNR_{dB} = 10 \log_{10} SNR$$

(a) Define bandwidth for a communication channel and then find the channel capacity.

(b) If there were no noise on the channel, how many signal levels would be needed to achieve the channel capacity found above in part (a). The Nyquist formula that relates these quantities is defined as:

$$C = 2B \log_2 M$$

where M is the signal (often voltage) levels.

-Analog and Digital Communication (EE-1, 2, 3)

FINAL EXAM (FALL 2015)

er 8, 2015

S: *Do not ask any questions from the invigilator; make your assumptions in case of confusion. labeled figures/block diagrams where required. There are total six (6) questions and 4 pages. Solve question together otherwise they will not be marked.*

EE323 - Analog and Digital Communications

Midterm Exam-2 Fall 2015

National University of Computer and Emerging Sciences, Lahore

Course Code: EE323 Semester: Fall 2015

November 7, 2015 Time: 60 minutes

Student Name: [signature] Roll number: l13-4432

Question-1 **10 Points**

A video signal for television transmission occupies a bandwidth 4.5MHz. These are the four possibilities for video transmission

1. Transmit the complete DSB with carrier
2. Transmit SSB with carrier
3. Transmit SSB (suppressed carrier)
4. Transmit VSB+Carrier that has 25% more bandwidth than SSB signal.

Copy and fill the table below on your answer sheets.

Transmission Option	Bandwidth Requirement	Significant Advantage(s)	Significant Disadvantage(s)	Preferable Detection Technique
DSB + Carrier				
SSB + Carrier				
SSB				
VSB + Carrier				

Question-2 **10 Points**

Draw the block diagram of DPCM (Differential Pulse Code Modulation) transmitter and receiver by using a 3^{rd} order linear predictor. Why we need a Quantizer block at the transmitter side? Describe with necessary equations.

Question-3 **10 Points**

A message signal $m(t)$, shown in figure below, modulates a carrier with frequency $f_c = 10KHz$ with $\Delta f = 1KHz$. The carrier amplitude is A. When the modulated signal is passed through the channel its amplitude is changed randomly. The received signal is demodulated by passing through different blocks as shown in the following block diagram. Sketch the output wave form, approximated, at points a, b, c, d, e. What are the values of f_c and Δf at point **c**?

...to determine when the 10,000th person has entered a fairground. Assume (a) a turnstile sensor connects to P3.5 and generates a pulse each time the turnstile is rotated and (b) a light is connected to P1.7 that is on when P1.7= 1 and off otherwise. Use interrupts.

National University of Computer & Emerging Sciences FAST

LAHORE

Maximum Marks: 100 **Total Time: 180 min**

EE323-Analog and Digital Communication (EE-1, 2, 3)

113-4432

FINAL EXAM (FALL 2015)

EE-3

Dated: December 8, 2015

***INSTRUCTIONS:** Do not ask any questions from the invigilator; make your assumptions in case of confusion. Make neat and labeled figures/block diagrams where required. There are total six (6) questions and 4 pages. Solve all the parts of a question together otherwise they will not be marked.*

Question # 1 [7+8+5 = 20 marks]

(a) Find the complex (exponential) Fourier series for the following signal $x(t)$ using Euler's formula. Then, find its Fourier transform and plot $X(\omega)$.

$$x(t) = \cos\omega_o t + \sin^2\omega_o t$$

(b) The spectrum of the message signal $m(t)$ is shown in Fig 1 below. To ensure communication privacy, this signal is applied to a system (known as **scrambler**) shown in Fig 2 below. Analyze the system and sketch the spectrum at points A, B, C and the output $x(t)$. Properly label the axis information.

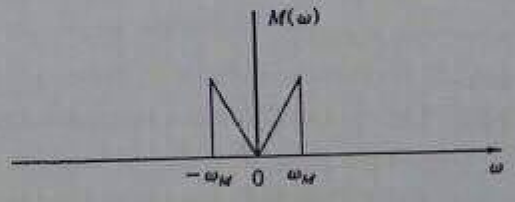


Figure 1 Spectrum of $m(t)$

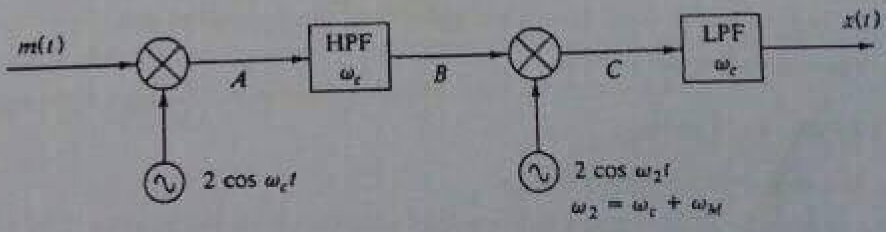


Figure 2 Scrambler

(c) Derive the envelope detector condition for SSB signal with carrier (SSB+C). For derivation, use LSB.